



Mathematical model for estimation of strength and weld bead geometry of dissimilar metal laser welds

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Abstract

The paper aims to propose mathematical model for predicting weld bead geometry and strength for dissimilar weld joints made of galvanized iron and stainless steel. The Analytical Hierarchical Process is used to identify significantly affecting process parameters on weld joints. The experiments are conducted to weld dissimilar metal sheets using Nd: YAG laser welding machine by considering the parameters obtained from central composite design matrix. The weld samples were prepared for measuring weld bead geometry and tested for tensile strength as per ASTM E8 standards. The responses are further processed for obtaining optimal condition using Genetic Algorithm and Fuzzy Genetic Optimization techniques are also compared with experimental results. The laser power, welding speed, and pulse duration are identified as significantly affecting parameters and laser power outranks among others. The proposed models predict results accurately greater than 85.29% in comparison with experimental results. The proposed model developed for predicting depth of penetration and ultimate tensile strength provides accurate results up to 8.03%, whereas 12.83% for bead width as compared to genetic algorithm and Fuzzy Genetic Optimization. The proposed model revealed 4.15% accurate results as compared to the experimental results. Mathematical models are useful for setting the process parameters and for obtaining the desired weld quality and characteristics in dissimilar metal weld joints.

Keywords Laser welding · Dissimilar metals · AHP · Buckingham's Pi theorem · Mathematical model

1 Introduction

The laser welding process is popularly used in automotive, aerospace, and electronics industries for joining complex parts due to its excellent weld characteristics and suitability for automation. The reliability of the weld joint depends on

the operating input process parameters and weld joint properties. Hence, it is essential to build a model to predict and optimize the parameters for achieving desired weld quality and weld joint material properties. The ideal combination of process parameters is chosen to achieve the desired quality of welds. In order to overcome this problem associated with the selection of appropriate process parameters and to obtain the desired output through developing mathematical models while also specifying the relationship between the input and output parameters. The several challenges to be faced during the welding of dissimilar metals are porosity, proper mixing of metals, Intermetallic compounds, uneven melting, spattering, cracking, formation of residual stresses, deformation of thin sheets, etc. In the last two decades, several designs of experiment techniques, optimization algorithms, and computational methods have been adapted and developed for welding applications.

The welding of stainless steel and galvanized iron sheets has great importance in the automobile and household

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utensils for the manufacturing of products. There is an ever-increasing demand for reliable and good-quality welds of dissimilar metals as these joints are susceptible to deformation and cracking due to their high coefficient of thermal expansion. The prime requirement of the automotive industry is to improve durability, reduce weight, and protect products from the environment. The base metal properties are varied to obtain optimal tensile strength, impact, and hardness properties under welding conditions and process parameters [1, 2]. Several authors have carried out the study on different types of materials to analyze the weldability of the combination of materials like Steel, Copper, Aluminum, Galvanized iron, Magnesium, and its alloys. Author also discusses the problems related to the welding of the same and dissimilar kinds of materials and their remedial solutions [3, 4]. Authors proposed a numerical simulation for welding of dissimilar metals to enhance the ductility. The results are evaluated based on deformation, maximum stress, and shear stresses [5].

The effect of process parameters on some mechanical properties of welded joints was statistically investigated and revealed that most of the process parameters are affecting on hardness and tensile strength [6]. Investigation of aluminum alloys, and low and high carbon steels to characterize the thermal. Mechanical, microstructural, and weld properties of dissimilar material weld joints to predict weld bead geometry and proposed a robust design to determine the appropriate controllable parameters. The optimization problems are addressed using the mean response model and transmitted variability model to fit both the controllable and noisy variables [7–9]. The Artificial Neural Network (ANN), Genetic Algorithm (GA), and Response Surface Methodology (RSM) are used to develop relationships between process parameters and establish the model for optimizing the responses. The input process parameters are optimized for the desired weld bead geometry using GA [10–12].

The Taguchi design and Analytical Hierarchical Process (AHP) methods are coupled to get optimum welding conditions for obtaining the desired weld bead geometry. It was observed that the welding speed is the most affecting parameter on the depth of penetration [13]. The welds were produced on 0.4 mm galvanized steel sheets under optimal parameters like laser power, welding speed, gap between weld samples, and focal position using response surface methodology [14]. The empirical models developed to predict weld bead geometry was created using plasma arc welding [15]. The investigation on the influence of laser welding process parameters on different types of efficiencies, weld pool geometry, and duty cycle of Nd: YAG Laser Welding [16–18]. The prediction of weld pool size and the area of vaporized zinc at the faying surface were computed and it was concluded that the failure load of galvanized DP980 steel joint strength increases with decreasing welding speed or increasing the laser power. The Taguchi method was used for optimized

results than existing signature welding protocols used by industrial firms [18, 19].

A hybrid Taguchi grey approach is used to obtain optimal parameters on laser beam welding of dissimilar metals revealing that the laser power significantly affects weld width [20]. The laser welding on austenitic SS steel sheet was carried out to study the effect of feed direction, wire distance, laser incident position, and laser incident angle on the wire melting mechanism, weld pool geometry, metal transfer modes, and heat distribution is analyzed using a high-speed camera and it was found that the fracture was always located in the lower thickness welds [21]. A review of dissimilar metal laser welding of electric vehicle battery connectors was carried out to overcome incompatible thermos physical properties like intermetallic changes, defects, and mechanical and electrical properties of joints [22]. The investigation was carried out to overcome the challenges like formation of brittleness, cracks, and corrosion in welds made of Steel, aluminum, and magnesium alloy combinations used metals in car bodies [23].

The study focused on providing a solution to the formation of porosity during laser welding of zinc-coated Steel to 304 SS Steel and suggested that the gap between two specimens leads to the escape of zinc vapor produced during welding [24]. A comprehensive review was carried out on the recent laser welding process, progressive development, and characterization of weld joints, monitoring effect of process parameters, length scale, and approaches. It was concluded that dissimilar metal welds always provide less strength than base metals due to the formation of intermetallic compounds [25, 26]. The author attempted to identify and analyze the welding 4.0 implementation issues through interactive structural modeling it reveals that intelligent decision-making is the driving barrier to overcoming resistance to change [27]. A hybrid optimization AI-based tool is built for selecting process parameters for obtaining better performance features of dissimilar metal laser welding [28]. A comprehensive review related to significance and effect of process parameters on Depth of Penetration (DOP) and Bead Width (BW) are given in Table 1.

It has been revealed from the literature review that many researchers carried out extensive research on the joining of dissimilar metals. However, limited literature was reported on the joining of galvanized iron and stainless steel sheets of less than 1.0 mm, and mathematical models for the prediction of strength and weld bead geometry are not available. Many researchers considered a limited number of process parameters and mathematical formulae are also not available for prediction of weld bead dimensions and weld strength of the sheets selected for the study. The success, failure, and quality of products created through welding depend on welding process parameters and the material used. This paper aims to

Table 1 The process parameters significance and effects on DoP and BW

Process Parameter	Significance	Interactions
Laser power [16–18, 25, 32, 34–36]	Significantly effects on processing speed and productivity	Increases in laser power increase the melting, and fusion Weld bead geometry and strength increases with increasing Laser power
Welding speed [16, 18, 24, 26, 32]	Quality and efficiency of the welding process	As speed increases HAZ and distortion decreases, whereas lower speed decreases the fusion zone Increase in welding speed leads to reduction in BW and DoP Welding speed is inversely proportional to DOP The increase in welding speed however results in the decrease of the maximum shear load An increase in the welding speed caused visible, irreversible structural changes in the fusion zone and HAZ
Pulse duration [18, 26, 34]	Increases power density at a certain energy	Increase in pulse duration results in heat input to the material and results in more melting Weld bead geometry and strength increases with increasing pulse duration
Focal length [26, 32, 35]	Affects significantly on BW and DoP	Optimal focal position ensures the balanced distribution of energy, control of HAZ and depth to width ratio For a specific value of Focal length results in optimal BW, DoP and UTS the weld strength tends to increase with increase in focal position
Beam diameter [18, 24, 29]	Controls the heat input and DoP	Increase in DoP fie a given welding speed DoP depends on welding speed and spot diameter DoP increases with for the smaller size of the spot for a specific power
Gas flow rate [30, 32, 37]	Increase in Shielding gas flow rate increases the size of BW & DoP	DOP and Strength increases with increase in gas flow rate
Frequency[16, 30]	Frequency increases power density	Increase in pulse duration results in heat input to the material and results in more melting
Beam angle[30, 32, 35, 37]	DoP and BW are controlled by varying beam angle	DoP increases from 0 to 90 degree and decreases thereafter, the behaviour BW is opposite to that of DoP At 90 degree beam angle provide optimal DoP and BW Incident beam angle significantly affect the shape and size of the weld bead geometry
Energy[16–18, 35]	Significantly effects on processing speed and productivity	Increases in energy input increase the melting, and fusion zone DOP and Strength of weld increases with increasing energy

identify the significantly affecting process parameters on dissimilar metal welds and also to propose mathematical models for predicting weld bead geometry dimensions and strength of the dissimilar metal laser weld joints made of less than 1.0 mm thickness.

2 Analytical hierarchical process

The analytic Hierarchy Process (AHP) is a systematic approach to making decisions based on knowledge, experience, intuition, and heuristics. The structure of the method

is based on a well-defined methodology derived from sound mathematical principles. It provides a formalized approach to making decisions for obtaining better solutions to complex problems. The AHP provides a means for decomposing the problem into a hierarchy of sub-problems which can further easily be comprehended and subjectively evaluated. The subjective evaluations are converted to numerical values and processed to rank each alternative on a numerical scale. The process parameters selected for the study are based on the feasible to vary parameters during experimentation. The experiments are conducted using Trumpf make 5004

Table 2 Parameters affecting on laser welding process

Sr. No	Parameters	Symbol	Sr. No	Parameters	Symbol
1	Laser power	P	5	Frequency	F
2	Welding speed	v	6	Gas flow rate	G
3	Pulse duration	T	7	Beam angle	Θ
4	Beam diameter	D	8	Energy	E

model, Nd: YAG laser welding machine and the corresponding process parameters used during welding are mentioned in Table 2.

The data regarding relative effect among process parameters was gathered from the experts and literature review. The numerical values in terms of gradation scale for comparison purposes are assigned based on the experience and judgment of the experts and literature review which strongly favor process parameters that are significantly affecting the welding process. A pair-wise comparison matrix is prepared and assigned relative weights to the pairs based on the effect of process parameters on responses and also interaction among them. The domination of one parameter over another was identified and subjective domination of the parameter is converted to numerical values from 1 to 9 which are given in Table 3. The computation of Eigen/priority value and consistency is done as per the algorithm suggested by Saaty's [37] is as follows.

1. Read the number of process variables (m)
2. Compute number of comparisons (N_c), $N_c = \frac{m(m-1)}{2}$
3. Initialize the comparison matrix as an identity matrix (C_{ms}), $C_{ms} = m \times m$
4. Read the upper triangular elements of the comparison matrix N_c
5. Assign the values to the lower triangular elements, $a_{ji} = \frac{1}{a_{ij}}$
6. Compute the sum of the columns of the comparison matrix (C_i), $C_i = \sum_{j=1}^m a_{jk}$
7. Normalize the elements of the comparison matrix, $a_{kj} = \frac{a_{kj}}{C_j}$
8. Compute the priority vector, $w_j = \frac{\sum_{l=1}^m a_{jl}}{m}$
9. Convert comparison matrix into diagonal matrix (d_m)
10. Compute the Eigenvector, $\lambda_{max} = d_m \times w_j$, $\lambda_{max} = d_m \times w_j$
11. Compare Eigenvectors obtained from MATLAB function (λ'_{max})
12. Check for consistency, $\frac{(\lambda_{max} - \lambda'_{max})}{\lambda_{max}} > 10\%$
13. Consistency index is $> 10\%$, go to step 4 and revise the relative scales
14. Else compute Consistency Index (CI), $CI = \frac{\lambda_{max} - m}{m-1}$

Table 3 Pair-wise relative grading

Description of parameters	Option	Numerical value/s
Two process parameter contribute equally to the objective	Equal	1
Experience and judgment slightly favor one parameter over another	Marginally Strong	3
Experience and judgment strongly favor one parameter over another	Strong	5
A parameter is favored very strongly over another, its dominance demonstrated in practice	Very Strong	7
The evidence favoring one parameter over another is of the highest possible order of affirmation	Extremely Strong	9
When compromise is needed	Intermediate values to reflect fuzzy inputs	2, 4, 6, 8
A reasonable assumption	Dominance of 2nd alternative as compared to first	Reciprocals

Eight possible process parameters are selected for the study, hence 28 comparisons are to be made based on relative grading scale. Table 4 prepared by considering the relative dominance of factors on each other, data from the literature review, and expert opinion through interviews. The relative comparisons of parameters are made and the graded values are used for pairwise grading. The upper diagonal matrix elements are allocated by referring to the highlighted grade values from Table 4, whereas lower diagonal elements are reciprocal of the upper diagonal elements. The priority weights of individual factors can be determined by the normalization method. The priority weights and weight percentages of each factor are calculated based on the sum of

Table 4 Relative grading of Process parameters

Parameters Factors	Extreme Favors	Very strong favor	Strongly favors	Slightly favors	Equal	Slightly favors	Strongly favors	Very strong favor	Extreme Favors	Factors
Laser power	9	7	5	3	1	3	5	7	9	Laser power
Laser power	9	7	5	3	1	3	5	7	9	Welding Speed
Laser power	9	7	5	3	1	3	5	7	9	Pulse Duration
Laser power	9	7	5	3	1	3	5	7	9	Beam Diameter
Laser power	9	7	5	3	1	3	5	7	9	Focal Position
Laser power	9	7	5	3	1	3	5	7	9	Gas Flow Rate
Laser power	9	7	5	3	1	3	5	7	9	Beam angle
Laser power	9	7	5	3	1	3	5	7	9	Energy
Welding Speed	9	7	5	3	1	3	5	7	9	Welding Speed
Welding Speed	9	7	5	3	1	3	5	7	9	Pulse Duration
Welding Speed	9	7	5	3	1	3	5	7	9	Beam Diameter
Welding Speed	9	7	5	3	1	3	5	7	9	Focal Position
Welding Speed	9	7	5	3	1	3	5	7	9	Gas Flow Rate
Welding Speed	9	7	5	3	1	3	5	7	9	Beam angle
Welding Speed	9	7	5	3	1	3	5	7	9	Energy
Pulse Duration	9	7	5	3	1	3	5	7	9	Pulse Duration
Pulse Duration	9	7	5	3	1	3	5	7	9	Beam Diameter
Pulse Duration	9	7	5	3	1	3	5	7	9	Focal Position
Pulse Duration	9	7	5	3	1	3	5	7	9	Gas Flow Rate
Pulse Duration	9	7	5	3	1	3	5	7	9	Beam angle
Pulse Duration	9	7	5	3	1	3	5	7	9	Energy
Beam Diameter	9	7	5	3	1	3	5	7	9	Beam Diameter
Beam Diameter	9	7	5	3	1	3	5	7	9	Focal Position
Beam Diameter	9	7	5	3	1	3	5	7	9	Gas Flow Rate
Beam Diameter	9	7	5	3	1	3	5	7	9	Beam angle
Beam Diameter	9	7	5	3	1	3	5	7	9	Energy
Focal Position	9	7	5	3	1	3	5	7	9	Focal Position
Focal Position	9	7	5	3	1	3	5	7	9	Gas Flow Rate
Focal Position	9	7	5	3	1	3	5	7	9	Beam angle
Focal Position	9	7	5	3	1	3	5	7	9	Energy
Gas Flow Rate	9	7	5	3	1	3	5	7	9	Gas Flow Rate
Gas Flow Rate	9	7	5	3	1	3	5	7	9	Beam angle
Gas Flow Rate	9	7	5	3	1	3	5	7	9	Energy
Beam angle	9	7	5	3	1	3	5	7	9	Beam angle
Beam angle	9	7	5	3	1	3	5	7	9	Energy
Energy	9	7	5	3	1	3	5	7	9	Energy

Table 5 Results of relative weights, priority vectors, $\lambda_{\max}$ and CI

Parameters	P	V	T	D	F	G	θ	E	Priority Vector
P	1.00	5.00	5.00	6.00	9.00	7.00	5.00	1.00	0.366
v	0.20	1.00	3.00	4.00	5.00	5.00	2.00	0.20	0.157
T	0.20	0.33	1.00	6.00	0.20	5.00	4.00	0.20	0.121
D	0.16	0.25	0.16	1.00	0.33	0.20	0.14	0.16	0.073
F	0.11	0.20	5.00	3.00	1.00	0.33	0.50	0.11	0.088
G	0.14	0.20	0.20	5.00	3.00	1.00	0.25	0.14	0.064
Θ	0.20	0.50	0.25	7.00	2.00	4.00	1.00	0.20	0.098
E	0.33	0.211	0.215	0.015	0.089	0.151	0.237	0.331	0.031

$$\lambda_{\max} = 11.6927 \text{ and } CI = 1.5$$

the priority weights. The relative weights, priority vectors, λ' and CI obtained are given in Table 5.

The four alternative criteria and six significant parameters led to the development of the matrix. The consistency ratio is less than 0.1. The resulting pair-wise comparison matrix is therefore consistent for all six criteria. The comparisons are acceptable, if Consistency Ratio (CR) is equal to 0. However, if CR is less than 0.1, the matrix contains erroneous judgments. The order of significantly affecting process parameters are given in the last column. The process

parameters ranked through AHP are arranged from most significantly affecting to the least are given in Table 6. The beam angle is ranked as the second significantly affecting process parameter but it is not considered for the study, the reasons are One, several authors have conducted experiments by varying beam angle, second angle variation facility was not available in the experimental set up used for welding. The laser power, welding speed, and pulse duration are ranked as the top three significantly affecting process parameters. Therefore, the experiments were planned using these three process parameters to study their effect on weld joint performance.

Table 6 Ranking of process parameters

Parameters	Percent weight	Rank
Power	36.606	1
Welding Speed	15.737	2
Pulse Duration	12.109	3
Beam Angle	9.822	4
Frequency	8.836	5
Beam Diameter	7.348	6
Gas Flow Rate	6.447	7
Energy Density	3.093	8

The experiments are conducted by varying laser power, welding speed, and pulse duration, and the pulse frequency and energy are also set to the desired constant values.

3 Experimental procedure

During the laser welding of dissimilar metals, the materials were mixing, and reaction with each other which results in a change of strength, hardness, ductility, brittleness, corrosion resistance, and weldability of engineering material must be known before use of particular material for engineering applications. Mechanical properties like ultimate tensile strength are essential to understand the behavior of the materials. The material properties of the 304L stainless steel and galvanized iron are considered for the study. These materials are popularly used in automotive and other allied industrial applications. The chemical properties of the material are given in Table 7 which is estimated using Spectro make, optical emission spectrometer (SPECTROMAXx). It is a stationary metal analyzer working on the principle of optical emission of vapors generated by electric arc spark. The galvanized iron sheet is coated with a 7 μm thick zinc layer.

The laser welding experiments are conducted on dissimilar metal sheets as per the L_{16} orthogonal array design matrix. Initially, the laser beam is focused at a focal distance of 150 mm. The focal length is above or below 150 mm the focus diameter changes, hence to confirm the size of the beam diameter, the laser beam is focused on thermal paper, and the burn-out portion is measured using an image analyzer. The dimensions obtained from the image analyzer are equal to 800 μm in diameter. The diameter of the beam selected is slightly greater than the thickness of the sheets since the shape of the HAZ region is oval in shape. With the selected beam diameter several trial experiments were conducted to identify the appropriate range of process parameters. The trial runs are conducted by varying the Laser Power (P), Welding Speed (V), and Pulse Duration (PD) independently from partial penetration to full Depth of Penetration (DOP)

and continue till the burning of sheets. The range of process parameters selected for experimentation is given in Table 8. The major challenges to be faced during conduction of experiments are selection of etchants, maintaining upset distance, gap between two specimens and fixing of specimen.

Weld specimens of size 150 mm $\times$ 60 mm are cut from the blank sheet with the help of a wire-cut electron discharge machine. Before welding, the workpiece samples are degreased and cleaned using 6–10% NaOH solution and then using 30% HNO_3 solution. The schematic experimental setup used for experimentation is shown in Fig. 1. The samples were welded with butt type of weld joints using Nd: YAG laser welding machine with wavelength of 1064 nm and designed to deliver maximum laser power of 4 kW. The oxygen welding produces some harmful gases that irritate the eyes, causing damage to the eye, and long-term exposure can cause permanent damage to human physiological functions. The rationale for using argon gas is to avoid harmfulness, distortion, and sputtering of the weld surface. Argon inert gas is recommended at a gas flow rate of 7 lit/min.

3.1 Sample Preparation

To examining weld bead geometry the sample sizes of 6 mm $\times$ 6 mm $\times$ 0.78 mm is cut from weld blanks using EDM. The cut specimens are placed in a hot mounting press chamber and then resin is added. A pressure of 25–28 MPa is applied to the resin-cut specimen together and heated up to a temperature of 177 $^{\circ}\text{C}$. It is then cooled to the atmospheric temperature by circulating cold water around the mold block. The typical hot-mounted samples are shown in Fig. 2(a). The hot-mounted samples are further processed for measurement of welding bead geometry and metallographic examination. Initially, samples are polished successively using 120, 240, 600, and 800 emery grits, followed by polishing with Leco DP-red oil-based lubricant. Finally, the weld joint is polished using a slurry of alumina, silica, or diamond with different grit sizes of 9 μm , 6 μm , 3 μm , and 1 μm to produce a scratch-free, mirror finish, free from smear, drag, or pull-outs and with minimal deformation remaining from the preparation process.

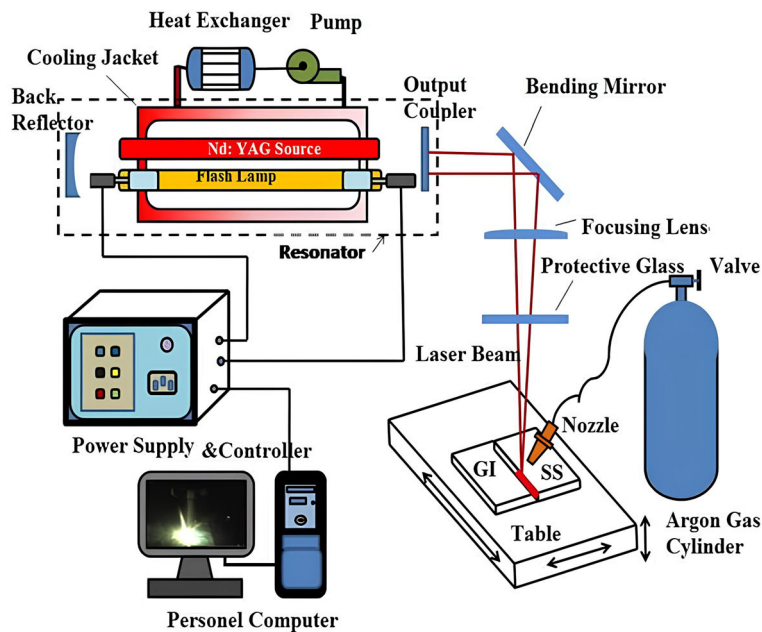
The polished samples are washed with distilled water and moisture is wiped out with cotton then samples are dried before etching. The carbon steel side of the weldment was etched in 4% Nital, and the rest of the regions of the weldment were etched in a solution containing 20 ml hydrochloric acid, 1.0 g sodium meta bi-sulfate, and 100 ml distilled water. Also, an electrolytic etching in 10% (w/o) oxalic acid was employed to reveal the features of the weld joint and its interfaces. The testing specimens are cut using water jet machining from weld plates according to ASTM E8 standard is shown in Fig. 2(b).

Table 7 Material Properties

Sheet Thickness	Elements	%C	%Mn	%Si	% Ni	%Cr	%Mo	%S	%P
0.5 mm	304L SS	0.03	1.09	0.38	0.03	17.9	8.99	0.66	0.45
	GI	0.06	0.40	0.02	0.01	0.01	0.03	0.001	0.05
0.78 mm	304L SS	0.03	1.09	0.38	0.03	17.9	8.99	0.66	0.45
	GI	0.06	0.40	0.02	0.01	0.01	0.03	0.002	0.05

Table 8 Process parameters and their levels

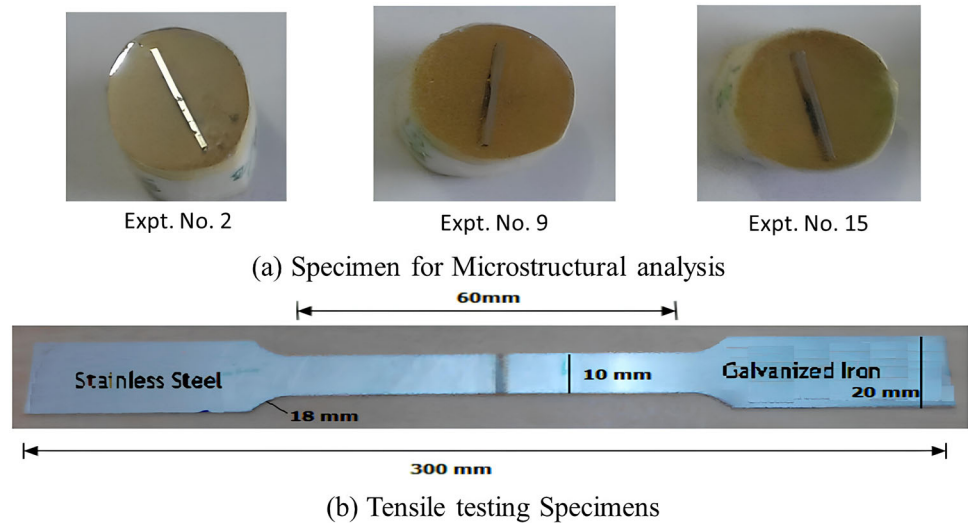
Process Parameters	Levels			
	1	2	3	4
Laser power (W)	1600	1800	2000	2200
Welding speed (mm/s)	2	4	5	6
Pulse duration (ms)	1	1.5	2	2.5
Focal Length (mm)	150			
Wavelength (μm)	1064			
Beam Angle (θ)	90	5		

**Nd:YAG Laser Welding Machine****Fig. 1** Schematic diagram of experimental setup

The cut specimens are tested for tensile strength on Universal Tensile Testing Machine (Fuel Instruments and Engineers (FIE) make, Model-UTN-40, Capacity 400kN). The mechanical properties of the welding joints are directly dependent on the geometrical form of weld bead geometry and its properties. It is an important physical characteristic of a weldment because several welding process parameters affect the weld bead geometry Fig. 3. Since Weldment characteristics like depth of penetration, bead width, bead length, and heat-affected zone region knowledge are extremely

important to characterize structural integrity. The dimensions of microstructure in the dissimilar metal weld joint have been obtained from Metatech make, metallurgical microscope. The weld joint images are captured at a magnification of 16X for the eye-piece lens and at 50X and 100X magnifications for the objective lens. The weld pool geometries were measured from all weld samples prepared. The average of three readings of weld bead geometry for each specimen is given in Table 9.

Fig. 2 Samples prepared for testing



WBG of 0.78 mm thick Samples

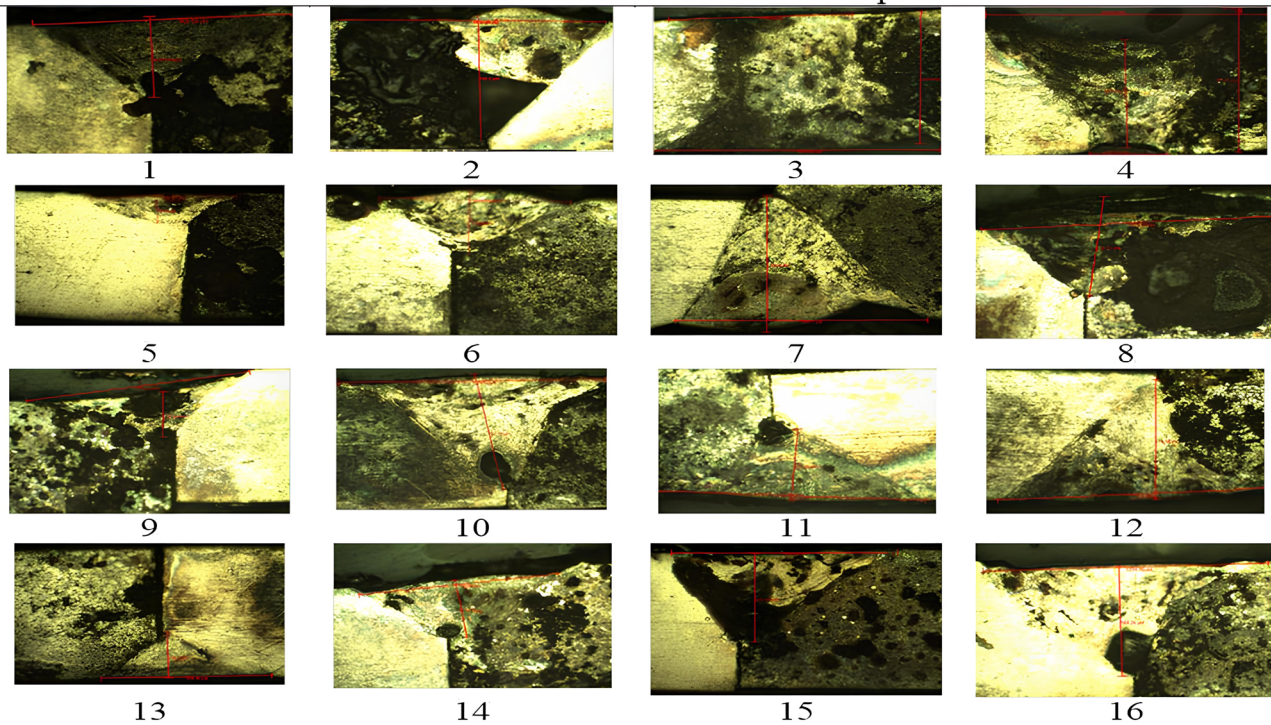


Fig. 3 Specimens prepared for metallographic inspection

4 Development of mathematical model

In order to develop three non-dimensional π terms, laser power, welding speed, frequency, energy, and pulse duration are selected from Table 8. The process parameters are selected by applying the following criteria, one, two variables should not have the same dimensions, second, process parameters should not form dimensionless parameters, third, none of them should be dimensionless and fourth is dependent variable should not be taken as a repeating variable.

The dimensional analysis (DA) is based on the hypothesis that the solution to the problem is expressible utilizing a dimensionally homogeneous equation in terms of specified process variables. The process variables and response parameters considered for experimentation are given in Table 10.

The principal use of DA is to characterize a phenomenon in terms of the relationships among dimensionless variables which are fewer in number than the original physical variables. This method helps reduce the complexity of experimental variables which affect the welding phenomenon.

Table 9 The process parameters and responses obtained from experimentation

Expt. no	Input parameter			Experiment Responses		
Parameters	P (w)	V (m/s)	t (s)	DOP μm	BW μm	UTS N/mm2
1	1600	0.003	0.0010	354	761	168
2	1800	0.003	0.0015	700	822	337
3	2000	0.003	0.0020	747	896	343
4	2200	0.003	0.0025	778	1081	263
5	1800	0.004	0.0010	593	695	175
6	1600	0.004	0.0015	341	812	342
7	2200	0.004	0.0020	741	941	290
8	2000	0.004	0.0025	530	1070	343
9	2000	0.005	0.0010	254	778	229
10	2200	0.005	0.0015	627	880	356
11	1600	0.005	0.0020	406	944	299
12	1800	0.005	0.0025	675	1043	315
13	2200	0.006	0.0010	254	685	285
14	2000	0.006	0.0015	326	787	309
15	1800	0.006	0.0020	471	842	329
16	1600	0.006	0.0025	589	1035	345

Table 10 Response and input process parameters

Sr. No	Variables	Symbol	SI Unit	Fundamental Dimensions
<i>Response parameters</i>				
1	Ultimate Tensile Strength	<i>UTS</i>	Kg m/ s ²	ML ⁻¹ T ⁻²
2	Bead Width	<i>BW</i>	m	L
3	Depth of Penetration	<i>DOP</i>	m	L
<i>Input parameters</i>				
1	Laser Power	<i>P</i>	Kg m ² / s ³	ML ² T ⁻³
2	Welding Speed	<i>v</i>	m/s	MT ⁻¹
3	Pulse Duration	<i>t</i>	s	T
4	Energy	<i>E</i>	Kg m ² /s ²	ML ² T ⁻²
5	Frequency	<i>f</i>	1/s	T ⁻¹

Many methods of DA are in practice, but this paper adopts Buckingham's π theorem to develop a mathematical model for estimating the weld strength, bead width, and depth of penetration.

5 Mathematical model for UTS

The ultimate tensile strength (UTS) of the material is described as the maximum stress that the can material withstand before breaking and is a function of P, v, t, f, and E. $UTS = \text{fun}(P, v, PD, E, BD, f, F, G)$ $UTS = \text{fun}(P, v, PD, E, BD, f, F, G)$ It can be represented in general form as given in Eq. (1).

$$(UTS, P, v, t, f, E) = k_1$$

$$(1) \quad \pi_1 = P^a v^b t^c S$$

where k_1 is a constant, six variables are selected for the study, out of these one is selected from the response and the remaining five from input process parameters. All selected variables are described in terms of three fundamental dimensions mass (M), length (L), and time (T). The total number of variables (p) = 6 and fundamental dimensions (q) = 3, therefore the number of dimensionless π terms is equal to $(p - q) = 3$.

$$(\pi_1 \pi_2 \pi_3) = k_1 \quad (2)$$

Application of Buckingham's π -theorem leads to several π -terms presented in terms of process parameters. The first π term is given in Eq. (3) in the form of power law.

$$(3) \quad \pi_1 = P^a v^b t^c S$$

Substituting the fundamental dimensions, equating the power components, and solving the power lead to a value of $a = -1$, $b = 3$, and $c = 2$. The expression for the first π -term is given in Eq. (4) and similarly π_2 , and π_3 are derived and given in Eq. (5).

$$\pi_1 = \frac{v^3 t^2 S}{P} \quad (4)$$

$$\pi_2 = \frac{E}{P t}, \pi_3 = f t \quad (5)$$

The π -terms do not provide the desired results directly, to get the results it is necessary to develop a relationship between these π -terms using a regression method. This method relates the dependent and one or more independent variables. The dimensionless parameters π_2 , and π_3 , are expressed in terms of π_1 (Eq. (6)) using power law.

$$\pi_1 = K \pi_2^{a_1} \pi_3^{a_2} \quad (6)$$

The Eq. (4) to Eq. (6) are represented in the form of Eq. (7) by taking the logarithm on both sides and assigning constant variables A, and B is given in Eq. (7).

$$Z = k_1 + a_1 A + a_2 B \quad (7)$$

where, $Z = \log(\pi_1 \pi_2 \pi_3)$, $k_1 = \log(K)$, $A = \log(\pi_2)$, $B = \log(\pi_3)$. The Z term contains A and B in the s dimensional coordinate system representing the hyperplane. The matrix elements are determined by summing and taking the product of A , B , and Z . The constant K , a_1 , and a_2 values are obtained by solving simultaneous equations (Eq. (8)).

$$\begin{bmatrix} -311.37 \\ 443.2 \\ -13.09 \end{bmatrix} = \begin{bmatrix} -22.75 & 32.45 & -0.99 \\ 0.675 & -0.99 & 0.11 \\ 139.39 & -198.60 & 5.89 \end{bmatrix} \times \begin{bmatrix} k_1 \\ a_1 \\ a_2 \end{bmatrix} \quad (8)$$

Further simplification of Eq. (7) is done by substituting k_1 , a_1 and a_2 which leads to the Ultimate Tensile Strength equation (Eq. (9)).

$$UTS = 7.747 \times 10^{-6} \frac{f^{1.048} v^{0.0633} E^{1.2393}}{P^{0.2343} t^{0.1495}} \quad (9)$$

6 Results and discussion

The mathematical models are developed for estimation of BW and DOP using Buckingham's π theorem resulting equations are given in Eq. (10 and 11).

$$BW = 7716.94 \frac{P^{0.1022} v^{0.0211} t^{0.398}}{f^{0.0516} E^{0.1022}} \quad (10)$$

$$DOP = 1177.1265 \frac{t^{0.057} E^{0.0329}}{v^{0.0614} f^{0.2025} P^{0.0329}} \quad (11)$$

The process parameters are ranked through AHP. The laser power, welding speed, and pulse duration are ranked as the top three significantly affecting process parameters, hence these three parameters are considered for conducting experiments. The BW and DOP are computed for both the metals independently using analytical Eq. (12).

$$DOP = \frac{(1 - R) P P D}{\pi r_0^2 \rho [L_m + C(T_m - T_o)]} \quad (12)$$

where R is the reflectivity, P is the power supplied (w), PD is the pulse duration (ms), r_0 is the beam radius (mm), ρ is the density of the material (kg/mm^3), L_m is the Latent heat of melting (K), C is the specific heat (J/kg K), T_m is the melting temperature (K) and T_o is the ambient temperature (K). A set of eight experiments on welding of dissimilar metals of 0.5 mm thickness was conducted as per the Taguchi L8 design of the experiment. The corresponding results obtained from experimentation are presented in Fig. 4. The depth of penetration of galvanized iron and stainless steel sheets are evaluated independently using empirical Eq. (12).

The DOP estimated using the empirical method for galvanized iron sheets is found to be greater than experimental whereas lower for GI-coated MS sheets. The experimental DOP results are found to be in close agreement with the majority of the results obtained from the empirical equation. As the present work focuses on the welding of dissimilar metals, hence it is assumed that both metals are equally melted and mixed during welding. The UTS results derived from the experimental and proposed methods are shown in Fig. 5. The proposed results of the mathematical model for experiment number 5 are found to be lower than the experimental results. The maximum deviation of 14.71% is observed, this is due to the lower heat input duration of 0.001 s. The experimental and proposed model results are found to be in close agreement with an average deviation of 4.83%.

Figure 6 depicts the BW results obtained from experimentation and the proposed model, the proposed model shows lower values for experiment number 4, 8, 12, and 16, whereas higher values for experiment number 5, 9, and 13 as compared to experimental results. The experiment number 15 result showed maximum deviation of 10.36% this is due to the welding operation being carried out at a laser power of 1800 W, welding speed of 0.006 m/s and heat supplied for the period of 0.0020 s. The experimental and proposed model results are found to be in close agreement with an average deviation of 4.39%.

Figure 7 shows the effect of process parameters on depth of penetration, revealing that the proposed model results are deviated from experimental results. The experiment number

Fig. 4 Comparison DOP results obtained from empirical and experimental method

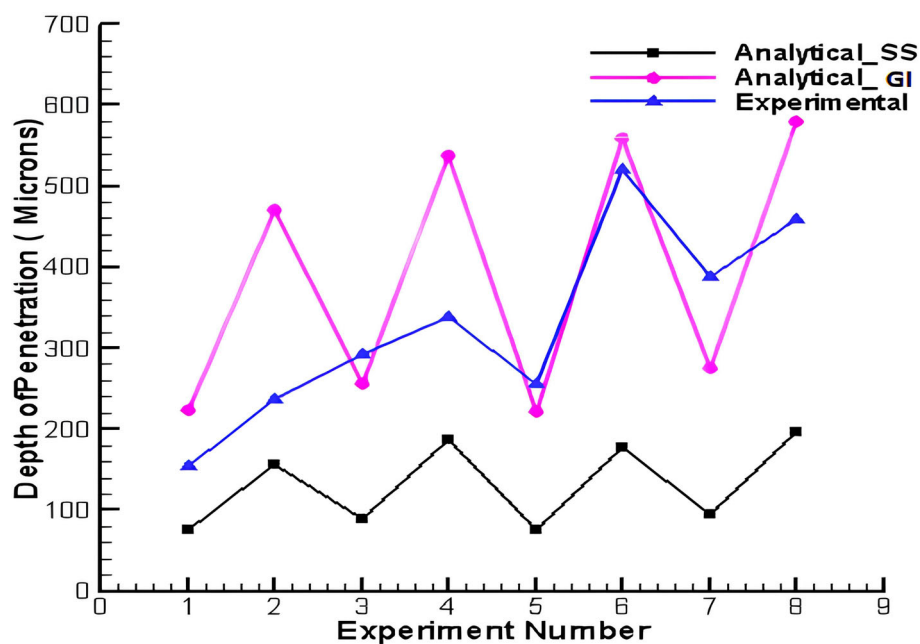


Fig. 5 UTS results obtained from analytical and experimental method

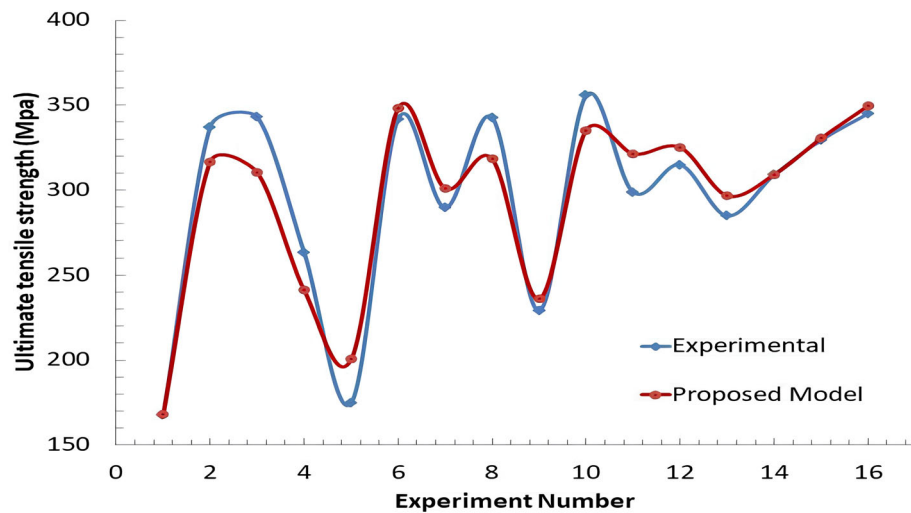


Fig. 6 Bead width results obtained from analytical and experimental method

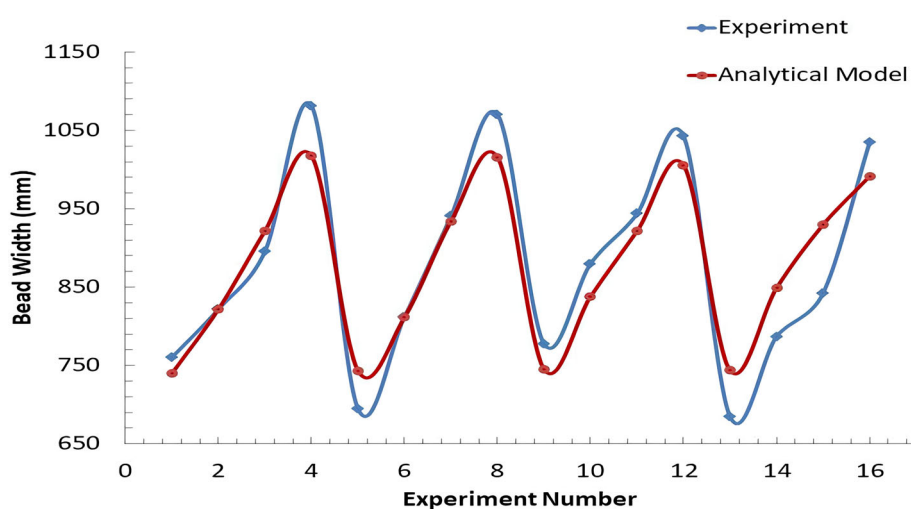
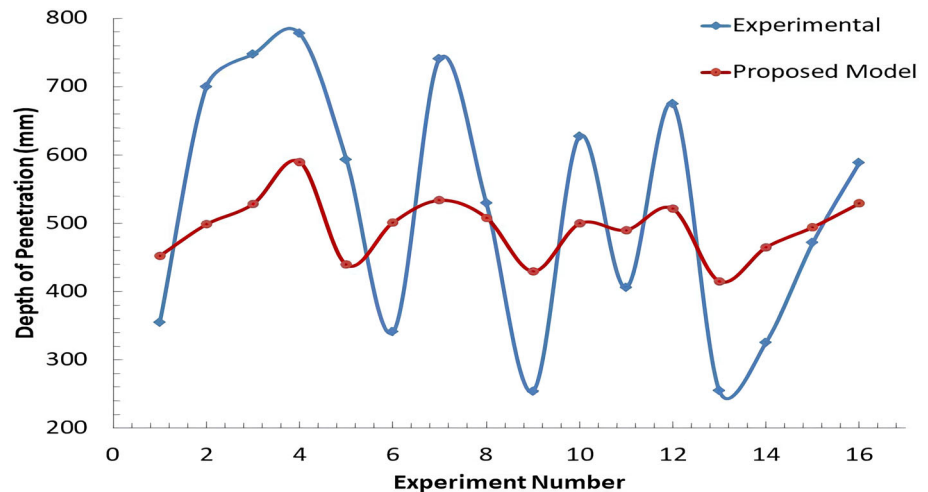


Fig. 7 Depth of Penetration from analytical and experimental method



8 and 15 are in better agreement with each other. This is due to a few significantly affecting process parameters considered for the development of proposed models. The results obtained from Eq. (12) are in close agreement with each other. The model considered for comparison is incorporated with all properties like reflectivity, beam radius, the density of the material, latent heat of melting, specific heat, melting temperature, and ambient temperature. The proposed model does not incorporate all the parameters mentioned in Table 1. Hence the results are deviated from experimental results. The development of a generalized model for welding of dissimilar metals is a very difficult task.

The FGO and GA techniques are used to evaluate the desired objective in terms of bead width, depth of penetration, and ultimate tensile strength. The results obtained after 5 lakh iterations from the FGO technique reveal that there is a decrease in the size of BW by 34.89% and an increase in strength of 35.64% as compared to the experimental results. The FGO predicts better results for DOP in comparison with the mathematical model and GA. Table 11 shows the comparison of results obtained from the genetic algorithm (GA), Fuzzy Genetic Optimization (FGO), proposed mathematical model, and experimental results. It reveals that the mathematical model and GA predict 4.15% less relative variation in BW and UTS respectively as compared to the experimental results.

The FGO predicts better results as compared to all other methods. The proposed mathematical model predicts better results than GA and FGO methods. The mathematical models proposed for the estimation of bead width provides accuracy up to 12.83%. The depth of penetration and ultimate tensile strength models predict accurately up to 8.03% and 8.83% respectively. The results obtained from the mathematical model and GA is in close agreement for parameters like bead width and ultimate tensile strength. However, the

GA predicts a 29% greater depth of penetration as compared to the mathematical model. The models developed are applicable in the manufacturing of automotive components, shipbuilding, aerospace industries for ensuring high-quality and durable weld joints. The mathematical models proposed in this work apply only to the combination of materials selected for the study and sheet thickness ranging from 0.5 mm to 1.0 mm. During mathematical modeling, only a few significantly affecting process parameters are incorporated.

7 Conclusions

The experiments are conducted using the process parameters ranked by AHP for laser welding of GI and 304L SS materials. The analytic hierarchical process is successfully implemented to rank significantly affecting process parameters. The laser power, welding speed, and pulse duration are identified as significantly affecting process parameters with laser power outranking others. The Proposed mathematical models were derived using the dimensional analysis technique for estimation of ultimate tensile strength, bead width, and depth of penetration in dissimilar metal welds of 0.78 mm. The proposed mathematical model predicts accuracy greater than 85.29% and 89.64% for UTS and BW respectively in comparison with experimental results. The depth of penetration results deviates less than 85% in comparison with the experimental results. The mathematical models proposed for the estimation of bead width, depth of penetration, and ultimate tensile strength provides accuracy of 12.83%, 8.03%, and 8.83% respectively as compared to FGO and GA. The GA predicts 29% better results of depth of penetration as compared to the mathematical model. This work can further be extended to develop a generalized analytical

Table 11 Comparison of results obtained from optimization techniques

Optimization Methods	Mean values of responses			Relative % error		
	BW	DOP	UTS	BW	DOP	UTS
Experimental (Confirmation test)	940.68	740.68	289.5	–	–	–
Mathematical model	934.04	633.55	300.03	0.63	14.35	3.81
Genetic Algorithm	925.88	494.38	301.63	01.59	33.06	4.15
Fuzzy Genetic Optimization	612.48	756.27	392.87	34.89	2.16	35.64

model for the prediction of DOP and weld strength of dissimilar metal laser welds made of different combinations of metals and thicknesses. The advanced tools can be used for developing mathematical models using mechanics theories.

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Declarations

Conflict of interest We confirm that there are no relevant financial or non-financial competing interests to report in this article.

References

- Seshu Kumar, G.S.V., Kumar, A., Rajesh, S., et al.: An experimental study and parameter optimization of FSW for welding dissimilar 6061 and 7075 Al alloys. *Int. J. Interact. Des. Manuf.* **17**, 215–223 (2023). <https://doi.org/10.1007/s12008-022-00913-1>
- Kim, J.K., Lim, H.S., Cho, J.H., Kim, C.H.: Bead on plate weldability of Al 5052 alloy using disk laser. *Achiev. Mater. Manuf. Eng.* **28**(2), 187–189 (2008)
- Mei, L., Chen, G., Jin, X., Zhang, Y., Wu, Q.: Research on laser welding of high strength of galvanized automobile steel sheets. *Opt. Lasers Eng.* **47**, 1117–1124 (2009). <https://doi.org/10.1016/j.optlaseng.2009.06.016>
- Iqbal, S., Muddassir, M.S., Gualini, A.R.: Dual beam method for laser welding of galvanized steel: Experimentation and prospects. *Opt. Laser Technol.* **42**, 93–98 (2010). <https://doi.org/10.1016/j.optlastec.2009.05.009>
- Poornima, C.L., Rao, C.S., Varma, N.: Numerical simulation on the laser beam welded UNS S32304 duplex steel and 304L stainless steel joints using ANSYS and response surface methodology. *Int. J. Interact. Des. Manuf.* (2024). <https://doi.org/10.1007/s12008-024-01860-9>
- Sinha, K., Kim, D.Y., Ceglarek, D.: Correlation analysis of the variation of weld seam and tensile strength in laser welding of galvanized sheets. *Opt. Lasers Eng.* **51**, 1143–1152 (2013). <https://doi.org/10.1016/j.optlaseng.2013.04.012>
- Ma, J., Kong, F., Kovacevic, R.: Finite-element thermal analysis of laser welding of galvanized high-strength steel in a zero-gap lap joint configuration and its experimental verification. *Mater. Des.* **36**, 348–358 (2012). <https://doi.org/10.1016/j.matdes.2011.11.027>
- Baghel, A., Sharma, C., Upadhyay, V., et al.: Optimization of process parameters for autogenous TIG welding of austenitic stainless-steel SS-304. *Int. J. Interact. Des. Manuf.* (2023). <https://doi.org/10.1007/s12008-023-01455-w>
- Harish Kumar D., Somireddy, A., Gururaj, K.: A review on critical aspects of 316 in austenitic stainless steel weldability. *Int. J. Mater. Sci. Appl.* **1**(1), 1–7 (2012). <https://doi.org/10.11648/j.ijmsa.20120101.11>
- Sathiyar, P., Paneerselvan, K., Sundararajan.: optimal design for laser beam butt welding process parameter using ANN GA for super authentic stainless steel. *Opt. Las. Technol.* **44**, 1905–1914 (2012). <https://doi.org/10.1016/j.optlastec.2012.01.025>
- Nagesh, D.S., Datta, G.L.: Genetic algorithm for optimization of welding variable for height to width ratio and application of ANN for prediction of bead geometry for TIG welding process. *Appl. Soft Comput.* **10**, 897–907 (2010). <https://doi.org/10.1016/j.asoc.2009.10.007>
- Shrimali, R., Kumar, M., Pandey, S., et al.: A robust Taguchi combined AHP approach for optimizing AISI 1023 low carbon steel weldments in the SAW process. *Int. J. Interact. Des. Manuf.* **17**, 1959–1977 (2023). <https://doi.org/10.1007/s12008-023-01307-7>
- Zaho, Y., Zhang, Y., Hu, W., Lai, X.: Optimization of laser welding of thin gage galvanized steel via response surface methodology. *Opt. Lasers Eng.* **50**, 1267–1273 (2012). <https://doi.org/10.1016/j.optlaseng.2012.03.010>
- Jiajun, Xu., Chen, C., Lei, T., Wang, W., Rong, Y.: Inhomogeneous thermal-mechanical analysis of 316L butt joint in laser welding. *Opt. Laser Technol.* **115**, 71–80 (2019). <https://doi.org/10.1016/j.optlastec.2019.02.012>
- Jahanzaib, M., Hussain, S., Wasim, A., et al.: Modeling of weld bead geometry on HSLA steel using response surface methodology. *Int. J. Adv. Manuf. Technol.* **89**, 2087–2098 (2017). <https://doi.org/10.1007/s00170-016-9213-0>
- Tadamalle, A.P., Reddy, Y.P., Ramjee, E., Reddy, K.V.K.: Influence of welding speed on melting efficiency of Nd: YAG laser welding. *Int. J. Adv. Prod. Eng. Manag.* **9**(3), 128–138 (2014). <https://doi.org/10.14743/apem2014.3.182>
- Tadamalle, A.P., Reddy, Y.P., Ramjee, E., Reddy, V.K.: Evaluation of Nd: YAG laser welding efficiencies for 304L stainless steel. *Procedia Mater. Sci.* **6**, 1731–1739 (2014). <https://doi.org/10.1016/j.mspro.2014.07.160>
- Tadamalle, A.P., Reddy, Y.P., Ramjee, E.: Influence of laser welding process parameters on weld pool geometry and duty cycle. *Int.*

- J. Adv. Prod. Eng. Manag. **8**(1), 52–60 (2013). <https://doi.org/10.14743/apem2013.1.153>
19. Achebo, J.I.: Optimization of GMAW protocols and parameters for improving weld strength quality applying the Taguchi method. In: Proceeding of the World Congress on Engineering, 1, WCE 2011, July 6–8 2011, London U.K. ISBN: 978–988–18210–6–5
 20. Thejasree, P., Natarajan, M., Khan, M.A., et al.: Application of a hybrid Taguchi grey approach for determining the optimal parameters on laser beam welding of dissimilar metals. *Int. J. Interact. Des. Manuf.* (2023). <https://doi.org/10.1007/s12008-023-01588-y>
 21. Xiaoyan, G., KaixuanZhu, S.W., Duan, Z., Zhu, L.: Effect of welding parameters on weld formation quality and tensile-shear property of laser welded SUS 301L stainless steel lap fillet weld. *J. Mater. Res. Technol.* **9**(3), 4840–4854 (2020)
 22. Sadeghian, A., Iqbal, N.: A review on dissimilar laser welding of steel-copper, steel-aluminum, aluminum-copper, & steel-nickel for electric vehicle battery manufacturing. *Opt. Laser Tech.* **146**, 107595 (2022). <https://doi.org/10.1016/j.optlastec.2021.107595>
 23. Karim, A., Park, Y.-D.: A review on welding of dissimilar metals in car body manufacturing. *J. Weld. Join.* **38**(1), 8–23 (2020). <https://doi.org/10.5781/JWJ.2020.38.1.1>
 24. Mohammad, G.R., et.al.: Weld bead profile of laser welding of dissimilar joint stainless steel. *IOP Conf. Ser. Mater. Sci. Eng.* **257**, 012072 (2017) <https://doi.org/10.10088/1757-899x/1/012072>
 25. Seyedeh, F.N., Anooshiravan, F., Hamid D.: A comprehensive review on recent laser beam welding process: geometrical, metallurgical, and mechanical characteristic modeling. *Int. J. Adv. Manuf. Tech.* **129**, 781–4828 (2023). <https://doi.org/10.1007/s00170-023-12536>
 26. Atmaca, E., S., & Kurt A.: The effect of welding speed and Focal length on mechanical characteristic fiber laser weld structures of DP600 Dual phase steel. *GU J. Sci. Part A* **8**(1), 146–156 (2021)
 27. Kumar, S., Sharma, K., Mishra, V., et al.: A structural model of Welding 4.0 implementation challenges. *Int. J. Interact. Des. Manuf.* **18**, 1243–1254 (2024). <https://doi.org/10.1007/s12008-023-01660-7>
 28. Thejasree, P., Narasimhamu, K.L., Natarajan, M., et al.: Generative modelling of laser beam welded Inconel 718 thin weldments using ANFIS based hybrid algorithm. *Int. J. Interact. Des. Manuf.* (2022). <https://doi.org/10.1007/s12008-022-00959-1>
 29. Diegel, C., et al.: Interaction between local shielding gas supply and laser spot size on spatter formation in laser beam welding of AISI 304. *Applied Science* **13**, 10507 (2023). <https://doi.org/10.3390/app131810507>
 30. Vyskoc, M., Domankova, M., et al.: The effect of Shielding gas flow rate on geometry, porosity. *Microstruct. Mech. Properties Laser Weld Joints* **28**(47), 1–10 (2020). <https://doi.org/10.2478/rput-2020-0021>
 31. Ghosh, P.S., Abhishek Sen, a, et al.: Progressive developments and challenges in dissimilar laser welding of steel to various other light alloys A comprehensive review. *Heliyon* **8**(11710), 1–15 (2022). <https://doi.org/10.1016/j.heliyon.2022.e11710>
 32. Gupta, S.K., et al.: Effect of welding speed on weld dead geometry and percentage dilution in gas metal arc welding of SS409L. *Mater. Today Proc.* **18**(7), 5032–5039 (2019). <https://doi.org/10.1016/j.matpr.2019.07.497>
 33. Radek, N., Pietraszek, J., Goroshko, A.: The impact of laser welding parameters on the mechanical properties of the weld. *AIP Conf. Proc.* **020025**, 1–7 (2018). <https://doi.org/10.1063/1.5056288>
 34. Gnanasekaran, S.: Effect of laser power on microstructure and tensile properties of pulsed Nd YAG laser beam welded AISI 301 austenitic stainless steel joints. *Mater. Today Proc.* **37**(2), 934–939 (2021)
 35. Matsumoto, N., et al.: Effects of laser focusing properties on weldability in high power fiber laser welding of thick high-strength steel plate. *J. Laser Appl.* **29**, 012003 (2017). <https://doi.org/10.2351/1.4966258>
 36. Palanisamy, S.: Effect of incident angle on weld microstructure and mechanical properties of laser beam welded nitronic 50 austenitic stainless steel joints. *Micron* **169**, 103457 (2023). <https://doi.org/10.1016/j.micron.2023.103457>
 37. Thomas, L.S.: Decision making with the analytic hierarchy process. *Int. J. Serv. Sci.* **1**(1), 83–98 (2008). <https://doi.org/10.1504/IJSSCI.2008.017590>

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